

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

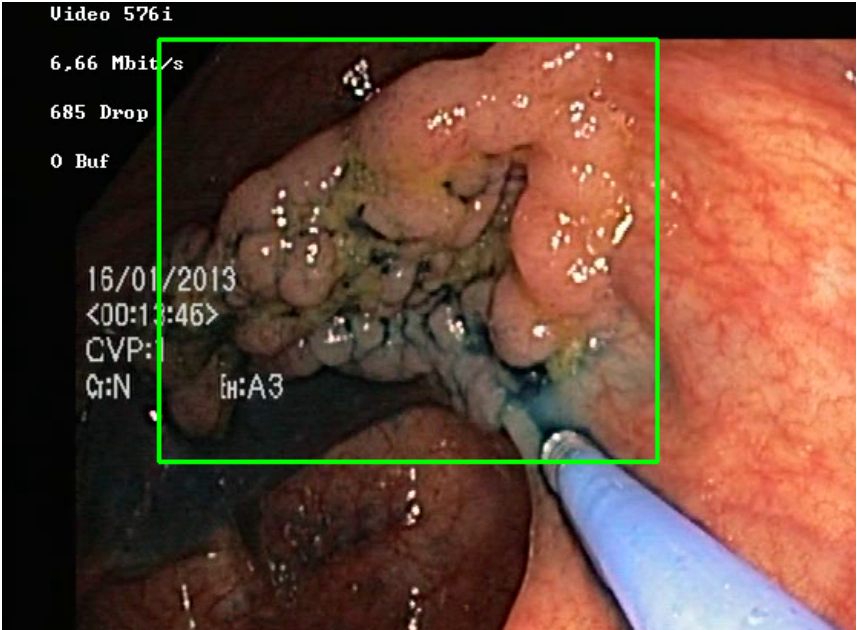
Video ID:	3fbc3436-372d-4665-b092-10b03b4dbd72
Total Frames:	2434
Frame Rate:	25.0 fps (assumed)
Duration:	97.4 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1356	✓
RT-DETR Detections	2258	✓
MedSAM2 Detections	2069	✓
Consensus (3-Model Agreement)	2	✓
Risk Classification	Count	Percentage
HIGH RISK	1	50.0%
MEDIUM RISK	0	0.0%
LOW RISK	1	50.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 370 frames)



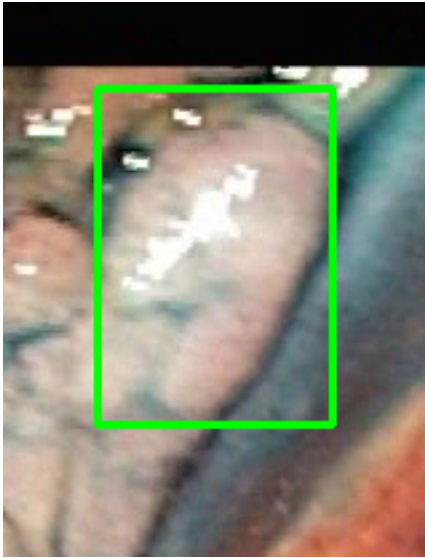
Feature	Value	Interpretation
Frame Index	2284	Frame 2284 of 2434
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.199	Color-based vascularization
Relative Radius	21.2%	Polyp size as % of frame diagonal
Texture Score	0.309	Surface morphology
Vessel Visibility	0.719	Vascularization indicator
Risk Tier	HIGH_RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended

Detection Confidence	83.5%	YOLO / RT-DETR / track confidence
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Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 684 frames)



Feature	Value	Interpretation
Frame Index	1527	Frame 1527 of 2434
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 92.0%
Redness Score	0.016	Color-based vascularization
Relative Radius	47.6%	Polyp size as % of frame diagonal
Texture Score	1.000	Surface morphology
Vessel Visibility	0.007	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	83.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 2 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (50.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 1 (50.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.